

AILamp

An expressive robotic lamp on a Jetson Nano

Group 8

Embedded Systems, Cyber-Physical Systems and
Robotics (INHN0018)

Technische Universität München, Campus Heilbronn

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Idea & goals

A lamp with five axes

01

The idea

A familiar desk object

Gestures express intent

Motion is our focus

02

Expressive motion

Inspired by ELEGNT

Recorded gestures

Five degrees of freedom

03

Our objective

Adapt LeLamp to Jetson

Replay selected motions

Verify each execution step

Contribution

On the LeLamp base (GPL-3.0)

01

Inherited

Mechanics and printed parts

MuJoCo model

Base runtime

Motion recordings

02

Our engineering

Jetson deployment

Replacement base design

Action control console

Home-relative playback

03

Demonstrated

Physical lamp movements

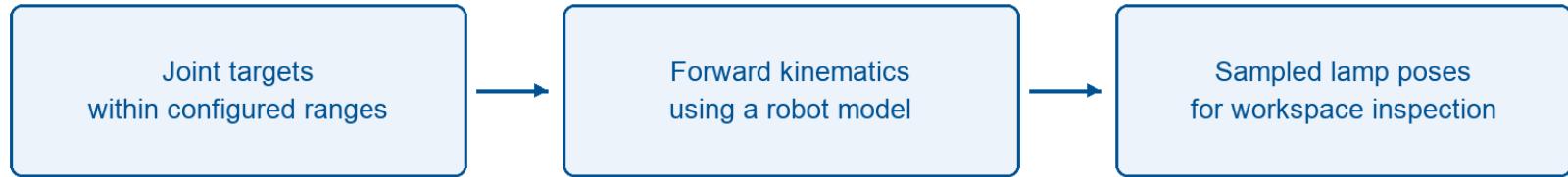
Saved home and poses

Operator-triggered actions

LeLamp (Human Computer Lab): an open-source expressive robot lamp.

Reachability analysis

Joint ranges and forward kinematics guide workspace inspection



Sampling explores the model workspace. It does not establish a complete reachable set.

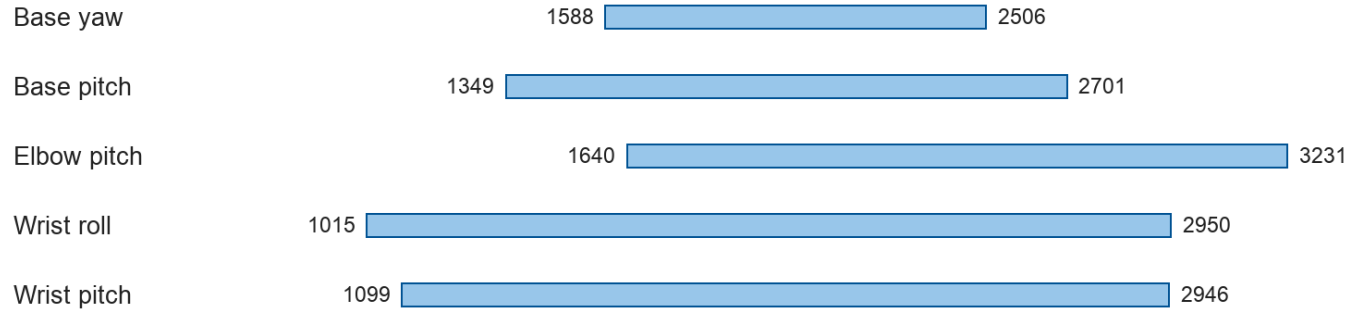
Physical checks must still address model mismatch, load and collisions.

Method illustration based on the LeLamp model. No certified workspace boundary.

Model sampling does not prove collision-free physical motion.

Motion constraints

Calibration, bounded targets and explicit motor enable



Raw encoder position: calibration snapshot, 8 September 2026

CSV targets use normalized units $[-100, 100]$, not degrees.

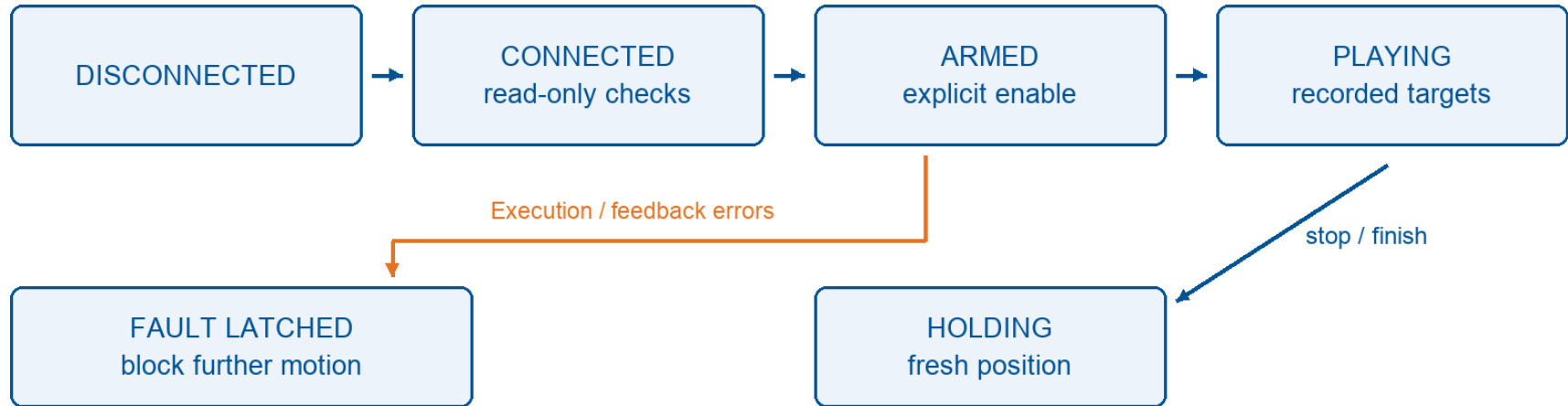
Stop reads the current position and holds it. Release switches torque off.

Position and communication faults block playback until checked.

Runtime checks reduce risk. Physical safety still requires validation.

Execution states

A conceptual view of connection, enable, playback and recovery



Conceptual workflow. Fault recovery needs inspection. Release requires physical support.

Motion functions

Save a home pose once, then select actions and return home

Function	Current implementation
Select	Browser action selection
Replay	Home-relative joint targets
Home	Return to saved pose
Stop	Read position and hold

Selected gestures

wake_up: attentive posture

nod: acknowledgment

headshake: disagreement

scanning: looking around

Browser controls running on Nano

One worker controls the motor bus

Voice integration is the next step

The console triggers movement. Saved home, rest and awake poses persist.

Base & assembly

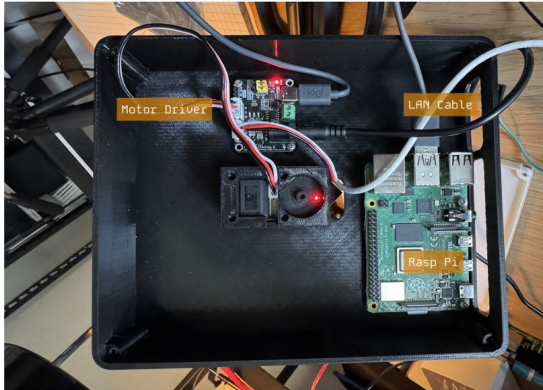
Replacement shell and cover for the Jetson configuration

The Jetson board needs a different electronics layout from the Pi baseline.

We designed a replacement shell and cover with ventilation and cable routes.

The video shows the assembled prototype with external wiring.

Upstream base, Raspberry Pi



LeLamp (Human Computer Lab, GPL-3.0)

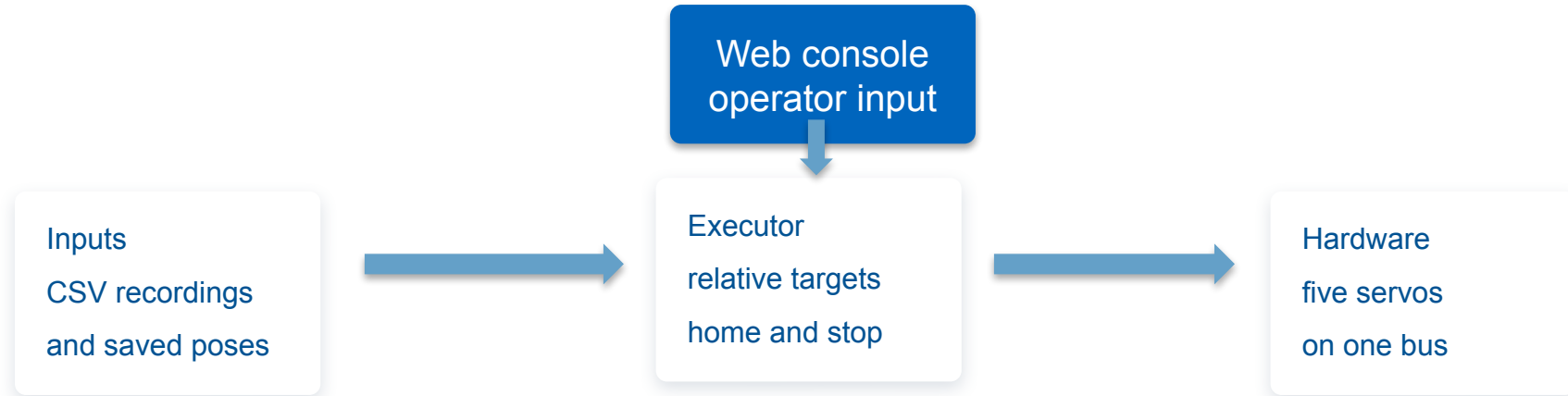
Our replacement shell and cover, Jetson



Rendered from the 3MF files in our repository

Software architecture

A dedicated motor execution path



Jetson Nano 4 GB runs the action console.

One worker sends joint targets and reads motor feedback.

Voice recognition is not connected to this execution path.

Action playback workflow

One saved home pose provides a consistent starting point



Returning home can interrupt playback. Incomplete returns remain visible.

Simulation & replay

Reference recordings and a user-defined starting posture

Recordings contain five normalized joint targets per frame at 30 Hz.

Current target = saved home + source frame - source first frame.



Wake-up



Curiosity



Scanning

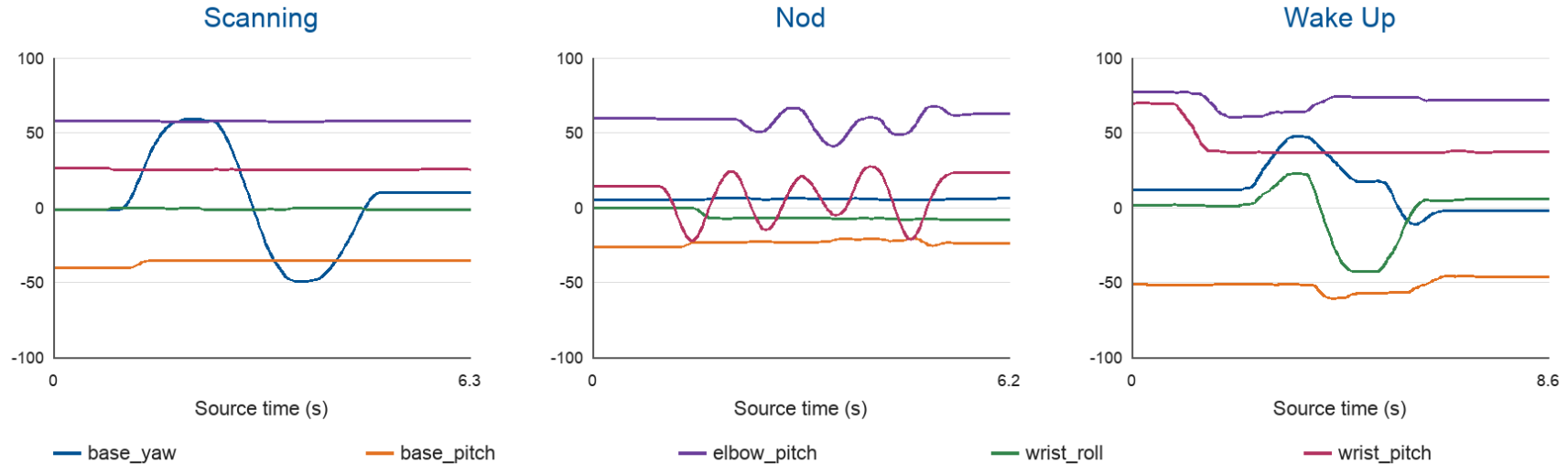
Frames from the supplied simulation video; LeLamp model (GPL-3.0).

Every converted frame is checked against calibration bounds before playback.

Evidence

Software checks and physical observations

13 source recordings checked. Physical gestures shown in the next slide.



Normalized CSV targets [-100, 100]. Source trajectories, not measured motion.

Demonstration

Physical demo → simulation animation | 2 min 18 sec



Voice-trigger scenario simulated; lamp movements triggered from the control console.

Course concepts in practice

Hybrid systems: sampled commands interact with continuous motor motion.

State management: connection, enable and active playback are distinct.

Actuator limits: calibration, dead band and load affect the result.

Shared I/O: five motors share a bus controlled by one worker.

Verification: command acceptance and physical arrival are different events.

Status & open work

Implemented and demonstrated
Nano action console deployed
Physical gesture video recorded
Home, rest and awake saved
Relative playback implemented

Next steps
Connect real voice commands
Reduce home-position error
Tune each physical gesture
Integrate vision and lighting

Conclusion

An operating prototype for expressive motion

We adapted LeLamp for Jetson Nano and built an action console.

The prototype performs physical movements and stores user-defined poses.

Next, we will connect voice input and improve repeatability.

Thank you. Questions welcome.

Sources & acknowledgements

- **LeLamp (Human Computer Lab):** mechanics, model, runtime, recordings, base photo. GPL-3.0.
- **Apple ML Research**, "ELEGNT", 2025.
- **Lee & Seshia**, Introduction to Embedded Systems.
- **Our fork:** GPL-3.0, provenance in NOTICE.md.

Backup: vision & voice

Voice-triggered interaction is the target, not the current trigger

The video illustrates a voice-triggered interaction scenario.

Actual motor actions are triggered from the control console.

Vision and voice prototypes exist in the wider repository.

Next integration: one recognized command calls one validated action.

Then test recognition errors, repeated requests and stop behavior.

Current demonstration: real motor motion with simulated voice triggering.

Backup: BOM & power

Current compute: Jetson Nano 4 GB and microSD storage.

Actuation: five ST3215 servos through a USB-connected servo driver.

The hardware design separates compute, motor and LED power domains.

Pico, LED matrix, camera and audio parts remain in the wider project BOM.

LED operation and speaker playback are not claimed in this demonstration.

Backup: test coverage

Reported software checks, with dates and boundaries

13 reference recordings contain 5,488 source frames.

Tests cover calibration identity, bounds, stopping and pose persistence.

New checks cover interrupted returns and relative trajectory conversion.

Recorded physical motion provides evidence beyond the simulated bus.

Position error, collisions and long-run reliability need further measurement.

```
Offline command: PYTHONPATH=ailamp_runtime python3 -m ailamp.motor_cli preflight
```

Backup: timeline

Early work: select the LeLamp platform and plan Jetson integration.

Build phase: assemble the mechanism and design the replacement base.

8-9 September: deploy the console and improve home and motion handling.

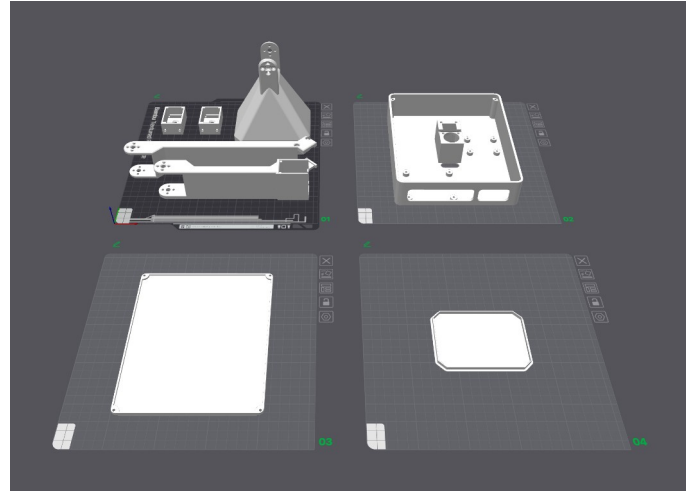
Presentation preparation: record physical gestures and integrate the video.

Next milestone: validated voice-to-action control on the physical lamp.

Backup: printable parts

Seven upstream parts, two replaced by ours

Our Shell and Cover replace LampBase and Cover. Arm, head, diffuser unchanged.



Print plate, LeLamp documentation (GPL-3.0)